Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/23/2007
PI Name:	Fox, George E Ph.D.		
Project Title:	Microorganisms in the Spacecraft Environment		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Environmental health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77204-5001	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	05/15/2004	**End Date:**	03/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 03/31/2007 per PI (7/07)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Willson, Richard C. (University of Houston)		
Grant/Contract No.:	NNJ04HF43G		
Performance Goal No.:			
Performance Goal Text:			
	The original project had two long-term objectives: (1) To develop general and specific microbial monitoring systems that utilize technologies that can be readily implemented in spacecraft. (2) To better understand the long-term effects of the space environment on representative bacteria. Three project tasks were to address these objectives. The primary task was to develop microbial monitoring technology that can be used on long term space missions. 16S ribosomal RNA, (16S rRNA), is the single most useful molecule for bacterial identification and therefore it was the target of choice in the monitoring work. In order to develop a general sentinel system for pathogens and other problematic organisms without prior knowledge of what organism might be encountered, we sought to take advantage of our earlier discovery that there are a large number of informative subsequences in 16S rRNA (“signature sequences”) whose presence in a unknown sample provide information that can		

Task Description:	be used to provide information on phylogenetic position (genetic affinity) or specific identity. For routine monitoring we proposed to look for the presence of specific genera and species as expected indicators of anticipated problems such as contaminated water or air. In a more sophisticated mode, we would seek to determine the genetic affinity of any problematic organism without prior knowledge of what it would be. Three approaches were investigated for this purposes; (A) microarrays of 16S rRNA targeted probes; (B) mass spectrometry and (C) solution hybridization with detection by fluorescence using molecular beacons. The essence of the second task was to conduct experiments with the objective of understanding how a low gravity/high background radiation environment affects bacteria over extended time periods. The third project task which was abandoned due to the early project termination was extension of the core monitoring technology to detection of microbial escape from an advanced life support system. The task addressed multiple critical questions in the HSR&T roadmap. This included needs for methodologies to detect and monitor microbial populations relating to human disease, and biofouling of equipment. The project also sought to obtain a better understanding of how bacteria respond to and adapt to the microgravity environment of space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A major project task is to develop microbial monitoring systems that can be used in space environments. Three alternative technologies, hybridization arrays, mass spectrometry, and solution hybridization with molecular beacons, are being examined. Each of these methods targets information carried by 16S rRNA and will require nucleic acid purification protocols. We are, thus, also examining simplified methods for isolating and purifying nucleic acid samples that can be used in space environments. At this stage, MALDI-TOF mass spectrometry appears to be the most promising approach. The key advantage mass spectrometry is that the core instrumentation is compatible with space flight and potentially has many uses. We initially made a detailed investigation of the feasibility of using the information obtained by digesting 16S rRNA from an unknown organism with a base specific nuclease such as ribonuclease T1 and identifying the mass of all the resulting fragments. Our analysis of this scenario, (see Zhang et al., BMC- Bioinformatics, 7: 117 [Epub ahead of print] (2006)) identified two key issues. The first was loss of information due to the inability of mass spectrometry to distinguish fragments with the same composition but different sequence (e.g. CACAAAG vs AAAACCG). It was found that fragments of length six and larger contained useful information and that enough of such fragments would be available to allow identifications to be made. The second difficulty is that the mass of cytosine and uracil are so close together that the relatively low resolution instruments that likely would be available during a space flight would not be able to distinguish them. We therefore have implemented an alternative experimental protocol in which the rRNA is first converted to DNA and then regenerated as RNA with a mass labeled analog of uracil Using this protocol we successfully identified five out of five bacteria in test runs (Jackson et al., manuscript in submitted) and have subsequently successfully used the approach with environmental samples.. In order to interpret the results one must compare the mass profile with that expected by known organisms. We therefore developed algorithms for doing this and these are described in a third paper (Jackson et al., 2006). In the array area, we have completed development of an evolutionary programming approach to design a universal array of less than 1000 15mer signature sequence probes. No organism is uniquely identified by a single probe. Instead, the design is such that any organism will test positive with at least 40 probes. The pattern obtained then serves to determine the genetic affinity of the unknown organism (idea is outlined in our recent paper-Chumakova et al., J. Biol. Phys & Chem., 5: 121-128 (2005)). With less than a thousand probes, this "signature chip" can in principle determine the genetic affinity of any bacterium. We have completed extensive computational validation of the concept, and developed an optimized set of signature sequences. These have been tested in small scale for methods development and optimization of spacer offset sequences, rRNA fragmentation methods, and labeling and analysis methods. Based on these results, a custom microarray was obtained from Combimatrix, and is currently being validated. A recent test of this chip with E. coli resulted in a positive identification. However, several probes either failed to hybridize or gave false positive signals. This was expected because the behavior of individual hybridization probes is notoriously unpredictable from sequence information alone. .These bad probes will need to be eliminated from the chip and replaced with alternatives that work. The third approach to microbial monitoring that is being pursued is the use of molecular beacons to identify specific bacteria for routine monitoring. Beacons carry a fluor that is initially quenched. When reaching their target in the 16S rRNA, the beacon probe hybridizes to the 16S rRNA resulting in exposure of the fluor and generation of a strong fluorescent signal. This solution hybridization approach would be advantageous because it would require minimal sample preparation and could be conducted with easily prepackaged reagents. The key problem is that the signal produced is insufficient for identification of small numbers of bacteria. Our proposed solution to this problem is to use a “beacon swarm” in which multiple probes would target the 16S rRNA molecule in multiple signature sites but emit fluorescence at a single wavelength. With ten targets a single molecule of target would produce an order of magnitude more fluorescence than is normally produced. To date, we have not been able to successfully demonstrate feasibility of the approach; possibly because of unintended interactions between the individual beacon probes. In order to perform microbial monitoring in the spacecraft environment using the methods discussed above (or alternative approaches developed by others) it will likely be necessary to have simplified procedures for obtaining and purifying nucleic acids samples. Work is continuing in two areas; the use of compaction agents to rapidly isolate RNA and DNA by selective precipitation and immobilized metal affinity chromatography (IMAC) for rapid purification. These areas were a major focus in year 1 and the substantial progress made then has now been finalized in year two with the result that three papers have now been published and two more have been submitted. The key new finding in year 2 is that nucleic acids can be eluted from IMAC columns by aqueous buffers. This is important because it eliminates the need to bring a potentially hazardous compound into the space environment. The second major project goal is to better understand the response of bacteria to the space environment. In order to do this, we are examining the growth of representative bacteria under simulated microgravity conditions using the NASA developed rotating wall bioreactor (HARV). An initial comparison of aerobic and anaerobic results revealed that there was no gene uniquely responding to these two growth conditions as would occur if the response was actually to gravity. We have completed our analysis of E. coli grown under aerobic conditions with results that show that its response is quite different from that of its close relative Salmonella. However, reviewers raised objections to the statistical analysis employed. We therefore have been collaborating on an independent parametric analysis of the data in collaboration with Dr. Yuriy Folfanov a biostatistician in our Computer Science Department in order to address the reviewer’s concerns.

	This analysis has been completed and a revised paper is being prepared for resubmission. In addition, we have been attempting to do long-term studies growth studies in the HARV to determine if the bacterium evolves a new response to the microgravity environment or alternatively if it simply relies on the short term adaptation on a long-term basis. These studies had until recently been hampered by difficulties in keeping the reactor contaminant free for extended periods. We have now developed a new protocol that has kept the reactor essentially contaminant free for up to six days.
Bibliography Type:	Description: (Last Updated: 10/31/2019)
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